

	(c)	e.g. satellite will already have some speed in the correct direction ...	B1	[1]
2	(a)	obeys the law $pV = \text{constant} \times T$	M1	[2]
		at all values of p , V and T	A1	
	(b)	$n = (2.9 \times 10^5 \times 3.1 \times 10^{-2}) / (8.31 \times 290)$	C1	
		$= 3.73 \text{ mol}$	A1	
	(c)	at new pressure, $n_n = 3.73 \times \frac{3.4}{2.9} \times \frac{290}{300}$		[3]
		$= 4.23 \text{ mol}$	C1	
		change = 0.50 mol	C1	
		number of strokes = $0.50 / 0.012 = 42$ (must round up for mark)	A1	
3	(a)	correct statement, words or symbols	B1	[1]